

Embedded Systems & IoT Handbook

Knowledge Base for AI Resume Analyzer

1. Introduction to Embedded Systems

An embedded system is a dedicated computing system designed to perform specific tasks efficiently. Unlike general-purpose computers, embedded systems are optimized for reliability, low power consumption, and real-time performance. Common applications include automobiles, medical devices, industrial automation, smart appliances, drones, and wearable technology.

2. Microcontrollers and Development Boards

Popular microcontrollers include Arduino, ESP32, STM32, PIC, and AVR. Arduino is ideal for beginners, ESP32 provides built-in Wi-Fi and Bluetooth for IoT applications, while STM32 is widely used in industrial products due to its high performance and extensive peripherals.

3. Communication Protocols

Embedded systems communicate using protocols such as UART, SPI, I2C, CAN, USB, and MQTT. UART enables serial communication, SPI offers high-speed peripheral communication, I2C allows multiple sensors to share two wires, and MQTT is commonly used in IoT cloud applications.

4. Sensors, Actuators and IoT

Sensors measure environmental parameters like temperature, humidity, pressure, motion and light. Actuators perform actions using motors, relays, LEDs or buzzers. A typical IoT architecture consists of sensors, microcontrollers, communication modules, cloud platforms and mobile/web applications for monitoring.

5. Resume-Worthy Embedded & IoT Projects

Examples include Smart Home Automation, Weather Monitoring Station, Smart Irrigation System, RFID Attendance System, Energy Monitoring Dashboard, Smart Parking Solution, Health Monitoring Device, and Industrial Machine Monitoring using ESP32, MQTT and cloud dashboards.

6. Best Practices & Career Roadmap

Master C/C++, Embedded C, Python, PCB basics, RTOS concepts, debugging tools, Git, and Linux. Learn ESP32, STM32, Raspberry Pi, communication protocols, cloud IoT platforms and edge AI. Build projects, document them on GitHub, and participate in hackathons to strengthen your resume.